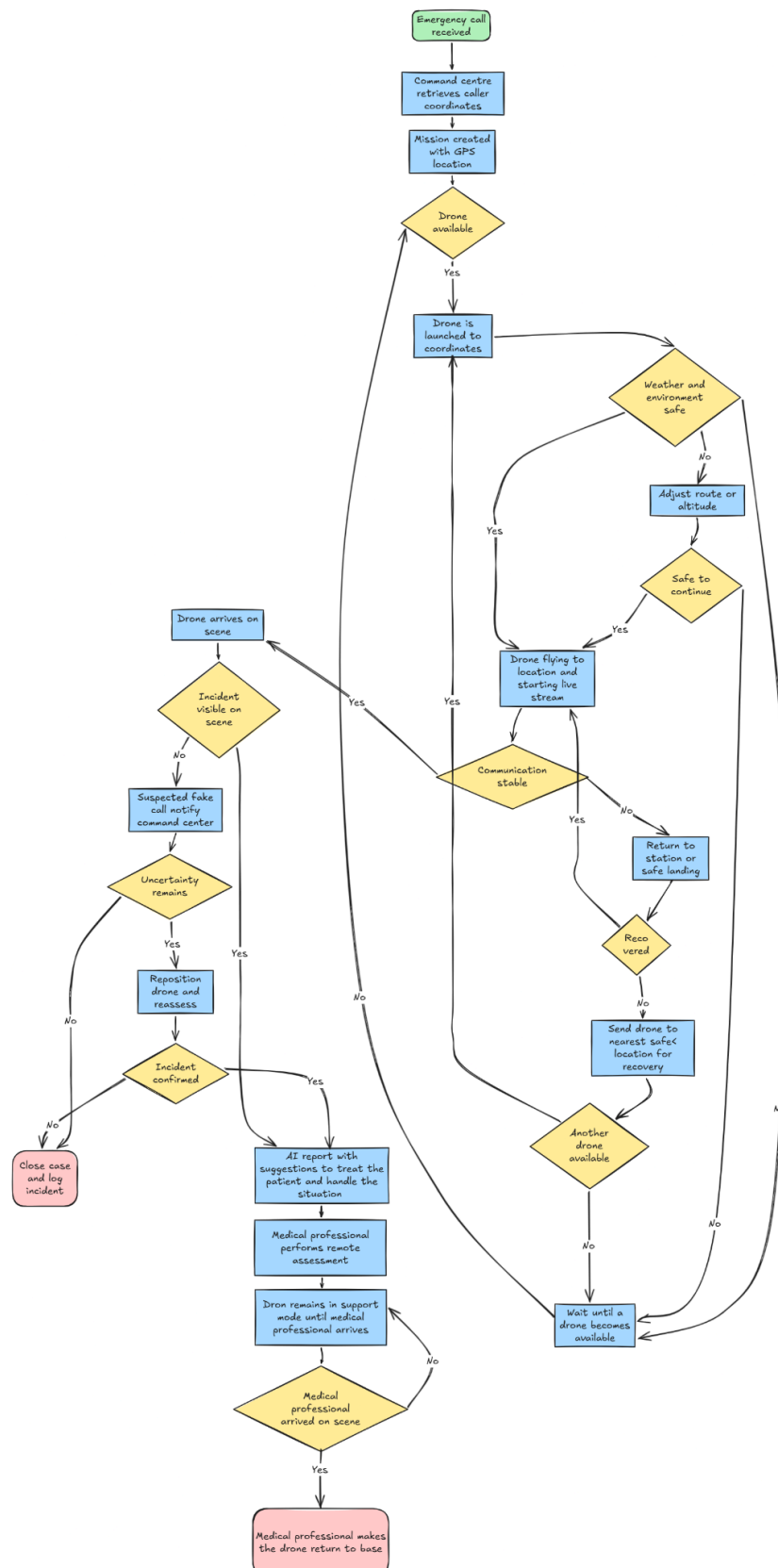


Crisis management protocol

- Flowchart to determine the deployment of drones



Protocol Structure – Disaster Response & Crisis Management

4.1 What to Do?

1. Receive alert of a medical incident
2. Deploy drone to incident location
3. Perform aerial assessment and live video streaming
4. AI report of the incident
5. Support on-site response until EMS arrival

4.2 When to Do What?

Phase 1 – Incident receipt

- Triggered by emergency call, security report.

Phase 2 – Drone Deployment (0–2 minutes)

- Immediate launch after receiving phone call

Phase 3 – Remote Assessment (2–5 minutes)

- Visual confirmation of:
 - Injury and number of the victims
 - Consciousness and mobility
 - Environmental risks

Phase 4 – EMS Coordination (parallel)

- Information relayed to paramedics and hospitals
- Route optimization provided

Phase 5 – Handover

- Drone remains in support mode until EMS confirms control

4.3 Who Is Responsible?

- **Command Center**

- Medical specialists
- Central communication hub
- **Medical Coordinator (EMS Lead)**
 - Interprets drone visuals medically
 - Determines response priority
 - Responsible for drone instructions
- **Paramedic Teams**
 - On-ground medical intervention
- **Hospitals**
 - Prepare for incoming patients based on early assessment

4.4 How to Communicate?

- **Visual Communication**
 - Live HD video feed to command center and medical coordinator
- **Public Communication**
 - Drone loudspeaker used when giving instructions
 - Clear, short, and calm instructions to bystanders
- **Data Handling**
 - Video data used strictly for medical and safety purposes
 - Compliance with privacy and data protection regulations

4.5 Which Procedures to Follow?

- Standardized medical assessment checklist for specialist in command centre.
- Escalation matrix for minor vs. critical incidents
- Emergency landing and fail-safe procedures
- AI learning after incident??

Communication Flow Overview

Alert → Drone → Command Center(professional) → Bystander → Hospital (if required)

Parallel: Controlled communication to bystanders/public when necessary.

Communication Plan

STEP 1: Receive Alert of a Medical Incident

Trigger Sources

- Emergency call (112)
- Security personnel report

Who Communicates?

- Reporting party → Professional in command Center
- AI→ intervening paramedics

How?

- Secure radio network
- Digital report notification

Information Shared:

- Exact location (GPS)
- Type of incident (if known)
- Number of victims (estimated)
- Immediate risks (crowd density, hazards)

- Severity of the injuries

Responsibility:

- Profesional in command center validates incident
- Profesional giving instruction to bystanders.
- AI makes an analysed report of the incident.

STEP 2: Deploy Drone to Incident Location**Who Communicates?**

- Emergency call sends drone automatically

How?

- By direct line

Information Shared:

- Analyse fastest route for responders

STEP 3: Perform Aerial Assessment & Live Video Streaming and support bystanders(on- site response).**Who Communicates?**

- Drone → Command Center

- Command Center → Medical profesional
- Live video feed shared with profesional
- Profesional medical→ bystanders

How?

- Secure live HD video stream
- Dedicated medical radio channel

Information Shared:

- Number of victims
- Severity of the injury
- Environmental hazards
- Crowd accessibility level
- Instructions to bystanders

STEP 4: AI Report of the Incident

Function:

AI system analyzes:

- Crowd density
- Victim movement
- Possible trauma indicators

- Route optimization for emergency responders

Who Receives AI Report?

- Command Center
- Emergency responders

How?

- Integrated incident dashboard
- Automated digital alert

Communication Protocol:

- AI output is a suggestion not a fact if it comes to the medical report.
- AI supports does not replace medical decision-making

Information Forwarded to respond forces:

- Severity category (Minor / Serious / Critical → hospital)
- Suggested access route
- Risk warnings